

Solutions for Scarcity

Diversification

Animals, especially humans, are not powerless in the face of food scarcity. The simplest strategy for scarcity of one resource is **diversification**. If one food is scarce, but another is unaffected by whatever caused the scarcity of the primary resource, then food scarcity may not be the result. You have likely experienced this yourself: You go to a grocery store, but they are out of whatever you intended to buy. In this circumstance; you don't go hungry, instead, you buy something else (maybe something less tasty, or more expensive, or not what you really wanted, but still filling). The same strategy is adopted by many animals: While they may prefer a specific food in a given season, they have alternatives that, while not preferred, are likely to be still available. The least preferred of these foods might be termed "**famine foods**" (also "fallback foods"), which are only eaten when no better food is available. Famine foods are often less tasty and less nutritious, or require extensive labor to process. One example is nuts: While it is relatively easy to crack pecans and chestnuts open and remove the edible seed (what we call the "nutmeat"), black walnuts and hickory nuts trap the seed in convoluted folds of the hard nutshell. Only smashing the nut into tiny pieces, and then laboriously picking out the nutmeat from the nutshell, will allow access to the edible portion. Modern industrial processes aid in this, and Native American learned to process such nuts more efficiently by boiling smashed nuts to produce nut oils, but we can still imagine that such labor-intensive resources were less likely to have been preferred foods in the past.

Diversification refers to growing multiple foods under multiple conditions to reduce the risk that any one hazard (e.g., hail, flood, or drought) will destroy the entire agricultural production in an area. Diversification strategies include diversifying what is grown (e.g., growing drought-tolerant varieties of crops alongside those that require more water); where it is grown (e.g., having some fields in a floodplain but others on higher land); and when it is grown (e.g., planting multiple crops of fast-growing plants in a year). It

also includes diversifying by raising animals as well as plants, and by incorporating wild foods into the diet.

Famine foods, or fallback foods, are those foods (usually wild plants and animals) that can be eaten but are not preferred; they are only eaten when none of the preferred, regularly eaten foods are available. These are often less nutritious or palatable, or much less efficient to prepare, requiring additional labor to make them edible.



Nuts that are easy to consume (from left, pecan and chestnut), and those that take significant effort to make edible (black walnut and hickory)

Source: Wikimedia Commons: [Pecan](#)

(https://commons.wikimedia.org/wiki/File:Carya_illinoensis_MHN

[T.BOT.2011.3.85.jpg](#)), [Chestnut](#)

([https://commons.wikimedia.org/wiki/File:Afgevallen_kastanje_van_een_Tamme_kastanje_\(Castanea_sativa\)_17-10-2021_\(d.j.b.\)_01.jpg](https://commons.wikimedia.org/wiki/File:Afgevallen_kastanje_van_een_Tamme_kastanje_(Castanea_sativa)_17-10-2021_(d.j.b.)_01.jpg)), [Black Walnut](#)

(https://commons.wikimedia.org/wiki/File:Black_walnut1.jpg),

[Hickory](#) (<https://commons.wikimedia.org/wiki/File:Pignut-hickory-nut.jpg>)

Overproduction and Storage

In the prior module on abundance, you read about the strategies that farmers use to store their crops from year to year to accommodate both regular times of scarcity, such as the winter, and the chance of loss of future years' harvests. For storage to be effective, more food must be produced than is needed in a single year (or, at least, in a single season). That food then needs to be either stored or traded for other durable items that can be exchanged for food at a future date, especially during a time of scarcity.